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Education

• **Jaypee Institute of Information Technology (JIIT), Noida**

2025–2029

Bachelor of Technology (B.Tech.) — Electronics & Communication Engineering

Projects

Sentinel: Interactive Cybersecurity Learning Platform

Tech Stack: Next.js, TypeScript, Tailwind CSS, Python, Gemini API, MITRE ATT&CK

- Built an interactive cybersecurity learning platform that guides users through simulated attack investigations instead of static tutorials.
- Implemented attack-flow visualizations, AI-assisted explanations, MITRE ATT&CK mapping, and downloadable investigation reports.
- Developed responsive investigation interfaces and reusable frontend components using Next.js, TypeScript, and Tailwind CSS.

Sentinel Edge: IoT Security & Edge Monitoring Platform

Tech Stack: ESP32, Python, FastAPI, Next.js, TypeScript, SQLite, MQTT, REST APIs

- Built an end-to-end IoT monitoring system using ESP32 and environmental sensors to collect and transmit live telemetry.
- Developed a FastAPI backend for device communication, data processing, REST APIs, anomaly detection, and alert generation.
- Built a web dashboard for live sensor readings, device health, historical trends, and security events using Next.js and TypeScript.

EdgeKart: Embedded Systems Marketplace

Tech Stack: React, TypeScript, Node.js, MongoDB, REST APIs, JWT, Tailwind CSS

- Built a full-stack marketplace focused on electronics components, embedded systems, and development hardware.
- Implemented authentication, product catalog, shopping cart, wishlist, and administrative workflows.
- Integrated MongoDB-backed application data with a responsive React/TypeScript frontend.

PortFlow: Customs Broker Onboarding Platform

Tech Stack: React, TypeScript, FastAPI, PostgreSQL, JWT, Tailwind CSS

- Built a production-inspired workflow platform for digitizing importer and customs-broker onboarding.
- Implemented customer management, document verification, compliance tracking, audit logging, and report-generation workflows.
- Connected a React/TypeScript frontend with a FastAPI and PostgreSQL backend.

Technical Skills

Languages: Python • TypeScript • JavaScript • SQL • Bash • C/C++

Frontend: React • Next.js • HTML • CSS • Tailwind CSS

Backend & Databases: FastAPI • Node.js • REST APIs • PostgreSQL • MongoDB • SQLite

Cybersecurity: Web Security • Network Security Fundamentals • OWASP Basics • MITRE ATT&CK • Linux

IoT & Embedded: ESP32 • MQTT • Sensor Integration • Embedded Systems

Tools: Git • GitHub • VS Code • Vercel • KiCad • OrCAD PSpice • AutoCAD

Libraries: NumPy • Pandas • Matplotlib • Framer Motion

Hackathons & Technical Activities

Climate Data Hackathon — JIIT

- Developed a Python-based analytics workflow for environmental datasets.
- Performed data preprocessing and visualization to identify and communicate dataset trends.
- Collaborated on analysis and presentation of findings.

UIDAI Data Hackathon 2026

- Designed Python-based analytical workflows for demographic datasets.
- Performed exploratory analysis and statistical visualization to communicate dataset insights.
- Presented findings through analytical dashboards and visualizations.

Ethical Hacking Workshop — IIT Delhi

- Practiced web-security concepts including XSS and SMB exploitation in a controlled workshop environment.
- Explored foundational vulnerability identification and exploitation techniques.

Leadership & Activities

Technical & Management Volunteer — AITRONICS Hub, JIIT

- Supported technical workshops and student events through participant coordination and event operations.
- Assisted with technical and organizational operations during the KRAKEN'X Hackathon.
- Contributed to event execution and participant management.